

Clinicopathologic and molecular correlates of overall survival in 173 surgically resected lung adenocarcinomas: A retrospective single-center cohort study

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Abstract

Background: Lung adenocarcinoma is the most common subtype of non-small cell lung cancer. Certain invasive histopathologic features, such as spread through air spaces (STAS), lymphovascular invasion (LVI), and visceral pleural invasion (VPI), are suspected to confer worse postoperative outcomes. The prognostic impact of these features and tumoral molecular profile on overall survival (OS) in resected lung adenocarcinoma remains to be fully characterized in contemporary cohorts. This study aimed to identify clinicopathologic and molecular correlates of OS in a large series of resected lung adenocarcinomas.

Methods: We retrospectively analyzed 173 patients with stage I-III lung adenocarcinoma who underwent curative-intent resection at a single academic center (2015-2019). Histologic subtype percentages, STAS, LVI, VPI, and pathologic staging were documented. Molecular testing for key driver mutations (EGFR, KRAS, ALK, ROS1, BRAF, MET exon 14) was performed. Survival outcomes were assessed using Kaplan-Meier estimates and compared by log-rank tests. Cox proportional hazards models identified independent prognostic factors.

Results: STAS was present in 24.3% of tumors and was associated with significantly worse 5-year OS (47% vs 72% for STAS-negative tumors; $p = 0.004$). VPI was identified in 54.3% of cases and also predicted inferior 5-year OS (54% vs 78% if no VPI; $p = 0.01$). LVI showed a trend toward worse survival (5-year OS 52% vs 71%; $p = 0.06$). Patients with any nodal metastasis had markedly reduced 5-year OS (N0: 75%, N1: 41%, N2: 24%; $p < 0.001$), and a higher lymph node ratio (LNR $> 25\%$) correlated with especially poor outcomes. On multivariable analysis, STAS, VPI, and nodal involvement (or higher LNR) emerged as independent adverse prognostic factors (all $p < 0.05$). KRAS-mutated tumors were associated with worse unadjusted survival (5-year OS approximately 30% vs 68% for others; $p = 0.003$), whereas EGFR-mutant tumors showed no significant survival difference. The cohort's molecular profile (22.5% EGFR-mutant, 6.4% KRAS-mutant) reflected regional demographics and influenced outcomes.

Conclusion: In this single-center surgical cohort, invasive histopathologic features STAS, VPI and nodal metastasis burden (+LNR) were strongly associated with OS. These factors retained prognostic significance independent of stage. KRAS mutations portended poorer survival, whereas EGFR mutations were not associated with worse outcomes in early-stage disease. Our findings highlight the importance of thorough pathologic assessment (STAS, LVI, VPI) and molecular characterization in resected lung adenocarcinoma to inform postoperative risk stratification and adjuvant therapy decisions.

Background

Lung adenocarcinoma is the most common histological subtype of non-small cell lung cancer (NSCLC), accounting for approximately 40-50% of cases. It is characterized by a heterogeneous mixture of histologic growth patterns (lepidic, acinar, papillary, micropapillary, solid), which form the basis of its classification in the World Health Organization (WHO) 5th edition of thoracic tumors. The latest WHO classification (2021) emphasizes detailed histologic subtyping (with percentage pattern quantification) and introduces a formal grading system for invasive lung adenocarcinomas. Notably, the 2021 WHO update also recognizes “spread through air spaces” as a distinct invasive phenomenon with adverse prognostic significance. STAS refers to tumor cells, specifically, the presence of micropapillary clusters, solid nests, or single cells floating within air spaces beyond the tumor’s edge. Since its description, STAS has been associated with aggressive disease behavior; multiple studies and meta-analyses have identified STAS as an independent predictor of higher recurrence rates and shorter survival in resected lung adenocarcinoma. The presence of STAS is now considered a negative prognostic factor in NSCLC, prompting investigations into its role in guiding surgical decisions of lobectomy vs sublobar resection.

Beyond STAS, other histopathologic factors are well-established to influence postoperative outcomes. LVI, tumor spread into lymphatic or blood vessels, facilitates micrometastatic dissemination and portends a higher risk of nodal metastasis and distant relapse. Accordingly, LVI has been consistently identified as

an independent adverse prognostic factor associated with decreased disease-free and overall survival. VPI, defined as tumor extension beyond the elastic layer of visceral pleura, is another critical risk factor. Tumors breaching the visceral pleura are upstaged in the Tumor-Node-Metastasis (TNM) system due to their worse prognosis under the 8th edition TNM classification, wherein any tumor ≤ 3 cm with VPI is classified as T2 (stage IB). Likewise, the presence of nodal metastases is a well-known adverse factor, and quantitative metrics of nodal involvement (LNR) may further stratify risk. Advances in surgical pathology and staging have increasingly highlighted these features in risk assessment for early-stage disease.

Molecular alterations also play a significant role in lung adenocarcinoma outcomes and therapy. A growing number of actionable driver mutations, including EGFR, ALK, ROS1, BRAF, MET exon 14 skipping, and recently KRAS G12C, have diagnostic and therapeutic implications. Current NCCN and ESMO international guidelines mandate routine testing for these biomarkers in all newly diagnosed advanced lung adenocarcinomas. The epidemiologic prevalence of driver mutations varies by patient ethnicity and geography. EGFR mutations are found in approximately 30-40% of Asian lung adenocarcinoma patients but only approximately 10-15% of those in Western countries. Conversely, KRAS mutations are far more frequent in Western cohorts (approximately 25-30%), compared to East Asian cohorts (roughly 8-10%).

These differences are reflected in large genomic studies, including The Cancer Genome Atlas (TCGA), which show KRAS as the predominant driver in North American and European patients, whereas EGFR is most common in Asian patients. Certain mutations carry prognostic weight as well; KRAS mutations, especially non-G12C variants historically lacking targeted therapy, have been associated with inferior survival in resected lung adenocarcinoma, whereas EGFR-mutant tumors may have distinct patterns of recurrence and responsiveness to EGFR inhibitors. For present-day diagnosticians, surgeons, and

clinicians guiding patient care, integrating molecular and clinicopathologic risk factors is essential for personalized management of early-stage lung cancer.

In this study, we present a single-institution retrospective analysis of 173 surgically resected lung adenocarcinomas (stages I-III) from an academic center. We examine the spectrum of histologic subtypes, invasion patterns (STAS, LVI, VPI), nodal burden, and molecular alterations (EGFR, KRAS, ALK, ROS1, BRAF, MET exon 14) and evaluate their associations with OS. Our findings are contextualized against contemporary benchmarks from the WHO classification, the 8th edition TNM staging criteria, and recent literature. The goal is to identify clinicopathologic and molecular correlates of outcome in resected lung adenocarcinoma, thereby informing prognostic assessments and highlighting features of potential clinical significance in routine pathology reports.

Methods

Study Design and Cohort

This study was a retrospective observational analysis of 173 consecutive patients who underwent curative-intent surgical resection for primary lung adenocarcinoma at our institution in Coimbra Portugal between 2015 and 2019. All patients had pathologic confirmation of lung adenocarcinoma. Clinical and pathologic data were extracted from institutional electronic medical records and original pathology reports. Collected variables included patient demographics, tumor characteristics, surgical procedure, and detailed pathologic findings. All cases were staged according to the AJCC/UICC, 8th edition TNM staging system.

Histologic and Molecular Characterization

Resected tumor specimens were reviewed by experienced thoracic pathologists. Histologic classification was performed according to WHO 5th edition (2021) criteria for lung tumors. Each tumor was classified

by its predominant histologic pattern (lepidic, acinar, papillary, micropapillary, solid), and comprehensive pattern percentages were documented in line with WHO recommendations. Tumor grade was assigned based on the predominant pattern (Grade 1: lepidic; Grade 2: acinar or papillary; Grade 3: solid or micropapillary), following the grading scheme proposed in the WHO 2021 classification. The presence or absence of specific invasive growth features, STAS, LVI, and VPI, was recorded for each case. STAS was defined as tumor cells beyond the main tumor border within air spaces of the lung parenchyma, per WHO definition. VPI was assessed using elastic tissue stains when necessary and defined as tumor extension beyond the elastic layer of the visceral pleura. LVI was noted when tumor cells were present within vascular or lymphatic channels.

Molecular testing for key driver mutations was performed on resection specimens, or pre-surgical biopsies if available, using standard clinically validated methods. Specifically, testing for EGFR mutations (exon 19 deletions, L858R, and other common variants) and MET exon 14 skipping was done by PCR-based assays, KRAS mutations were assessed by PCR or sequencing, and ALK and ROS1 rearrangements were evaluated by immunohistochemistry and confirmatory FISH. BRAF V600E mutation status was determined by targeted sequencing. All molecular tests were done in CAP/CLIA-approved laboratories with appropriate controls.

Nodal Metrics

The extent of nodal dissection and nodal metastasis was recorded. The total number of lymph nodes examined and the number of positive (metastatic) lymph nodes were documented for each case. From these, the LNR was calculated as the proportion of positive nodes out of total nodes examined. Patients were classified by nodal status (pN0, pN1, or pN2) according to TNM 8th edition criteria. LNR was analyzed both as a continuous variable and in categorical tiers, for example, low vs high LNR, with a threshold like 25% for the “high LNR” category, for prognostic impact.

Survival Analysis

OS was defined as the time from the date of surgery to the date of death or last follow-up. Patients still alive at last follow-up were censored. Median follow-up time was calculated among censored patients. Survival curves were estimated using the Kaplan-Meier method, and differences between groups (e.g. presence vs absence of a feature) were compared using the log-rank test. Five-year OS rates were derived from the Kaplan-Meier estimates. Where appropriate, survival outcomes were stratified by clinicopathologic features (e.g. STAS vs no STAS, VPI vs no VPI, etc.) and by molecular subsets.

Statistical Analysis

Univariate associations between individual factors and survival were evaluated with Cox proportional hazards regression to estimate hazard ratios (HRs) and 95% confidence intervals (CIs). A multivariable Cox regression model was constructed to identify independent prognostic factors for OS. Factors included in the multivariable model were those considered clinically important or found to be significant in univariate analysis ($p < 0.05$). The final model included variables such as STAS, LVI, VPI, nodal status or LNR, tumor size, and other covariates as needed to adjust for potential confounding. HRs > 1 indicated an increased risk of death associated with that variable. All statistical tests were two-sided, with a significance threshold of $p < 0.05$. Statistical analyses were conducted using SPSS (version 26) and R software (version 4.3.2).

Results

Cohort Characteristics

A total of 173 patients met the inclusion criteria, with baseline characteristics summarized in Table 1. The mean age was 66.6 years (range 38-88), and 63.0% were female. Most patients (approximately 90%) had a history of never smoking or remote light smoking, reflective of the high female proportion and the known epidemiology of lung adenocarcinoma in never-smokers. All patients underwent an anatomic lung resection (158 lobectomies and 15 sublobar resections) with curative intent. Final pathologic stages (per

8th edition TNM) were: stage I in 118 patients (68.2%), stage II in 28 (16.2%), and stage III in 27 (15.6%). Table 1 summarizes the demographic, histologic, molecular, and clinical characteristics of the surgical cohort (n = 173), alongside reference values drawn from WHO 5th edition classification and large-scale datasets (from the TCGA) where available. Notable deviations from reference frequencies are included for context.

Table 1. Patient characteristics and reference cohorts.[‡]

Characteristic	Study Cohort	Reference %	Notable Deviation
Demographics			
Total patients (n)	173	—	—
Age, mean (years)	66.6	—	—
Sex: Female	109 (63.0%)	—	—
Sex: Male	64 (37.0%)	—	—
Histologic Subtype			
Acinar	82 (47.4%)	40–50%	—
Lepidic	24 (13.9%)	10–15%	—
Papillary	17 (9.8%)	10–15%	—
Micropapillary	16 (9.2%)	5–10%	—
Solid	10 (5.8%)	5–10%	—
Other/unspecified	24 (13.9%)	< 5%	High unspecified rate
Pathologic Features			
STAS present	42 (24.3%)	15–20%	Above typical range
LVI	47 (27.2%)	—	—
VPI	94 (54.3%)	—	—
Molecular Alterations			
EGFR	39 (22.5%)	15–35%	—
KRAS	11 (6.4%)	20–35%	Lower than expected
ALK	5 (2.9%)	3–7%	—
ROS1	2 (1.2%)	1–2%	—
BRAF	1 (0.6%)	1–3%	—
MET exon 14	2 (1.2%)	1–4%	—
Nodal Assessment			
Median lymph nodes	15	—	—
Median metastatic nodes	0	—	—
Median nodal ratio	0	—	—
Survival Outcomes			
Alive (at last follow-up)	99 (57.2%)	—	—
Deceased	74 (42.8%)	—	—
Median OS (days)	2238	—	—

[‡]Reference ranges are derived from the WHO classification and TCGA lung adenocarcinoma data. Abbreviations: STAS - spread through air spaces; LVI - lymphovascular invasion; VPI - visceral pleural invasion; EGFR - epidermal growth factor receptor; KRAS - Kirsten rat sarcoma viral oncogene; ALK- anaplastic lymphoma kinase; ROS1 - ROS proto-oncogene 1; BRAF - B-Raf proto-oncogene; MET - mesenchymal-epithelial transition factor; OS - overall survival.

Histopathologically, the tumors demonstrated a diverse mix of adenocarcinoma subtypes. The predominant histologic pattern was acinar in 82 cases (47.4%), followed by lepidic in 24 (13.9%), papillary in 17 (9.8%), micropapillary in 16 (9.2%), and solid in 10 cases (5.8%). An additional 24 tumors (13.9%) were classified as “other/unspecified” in the original pathology reports, typically indicating

ambiguous features or no single dominant pattern. On comprehensive re-review under current WHO criteria, many of these “unspecified” cases likely represent mixtures without a clear predominant pattern.

Our subtype distribution falls within expected ranges from larger reference series; for example, acinar is generally the most common pattern worldwide (around 40-50% of cases), whereas micropapillary and solid patterns together comprise < 20% in most cohorts. The relatively high “unspecified” rate (13.9% in our series, versus < 5% in reference datasets) highlights the importance of thorough pattern reporting; on re-examination, many of the unspecified cases in our study showed mixed patterns without a single clear predominant type.

Invasive histologic features were prevalent in this resected cohort. STAS was identified in 42 tumors (24.3%), LVI in 47 (27.2%), and VPI (PL1 or PL2) in 94 tumors (54.3%). Notably, our observed STAS incidence of 24.3% is slightly above historical averages. STAS has been reported in roughly 15-20% of resected lung adenocarcinomas in prior studies, though rates vary widely, up to approximately 70% in some series, depending on how aggressively STAS is sought and the cohort’s variable stages.

The high VPI frequency (54.3%) reflects that over half of the tumors had invaded the visceral pleura (T2 by definition); this is partly attributable to many tumors being peripherally located and relatively large, mean invasive size 2.8 cm. There was a strong association between STAS and other invasive features; for instance, STAS-positive tumors more often had concomitant VPI and LVI than STAS-negative tumors (each comparison $p < 0.01$), consistent with the inter-correlated nature of aggressive pathologic factors reported in the literature.

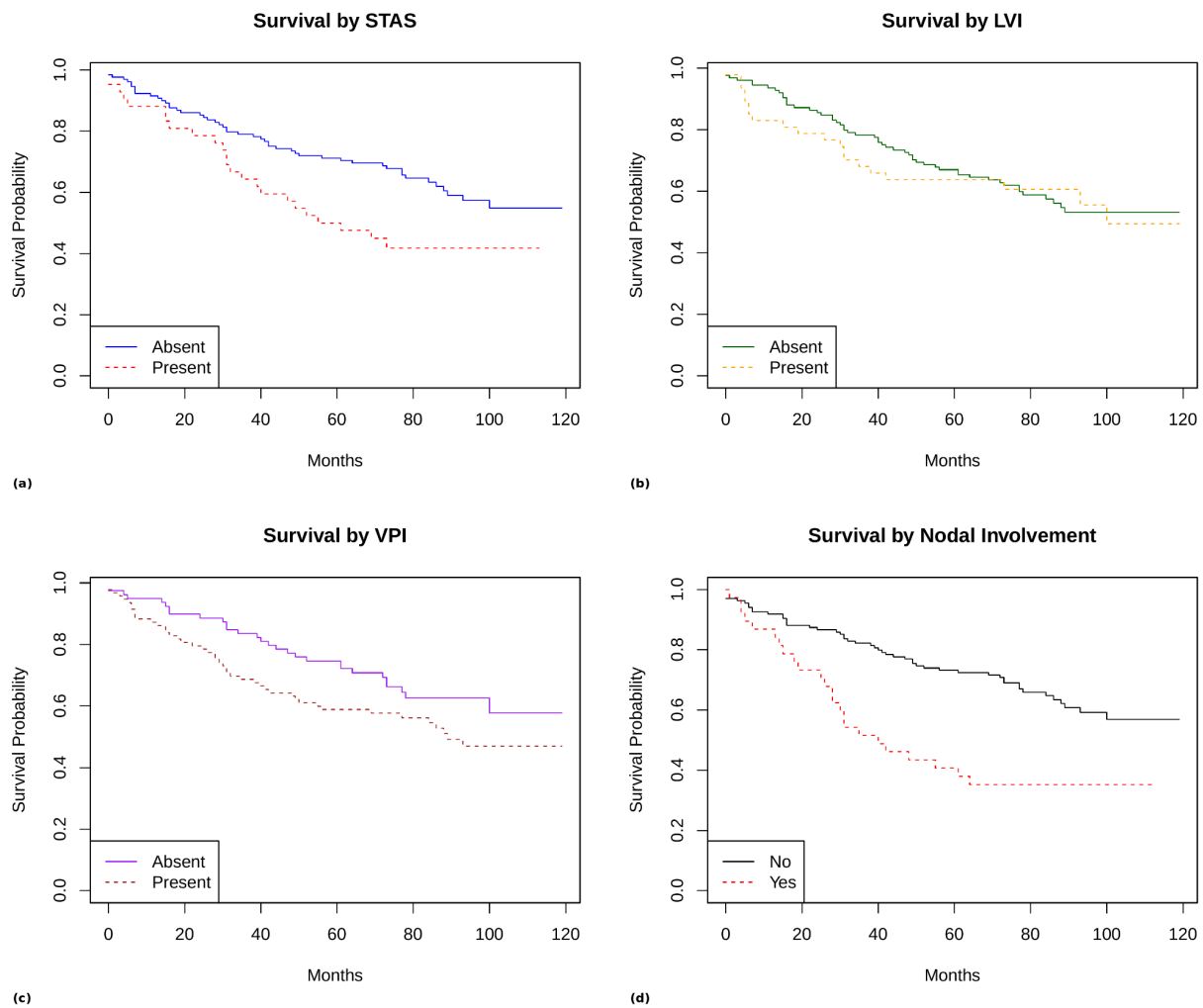


Figure 1. Kaplan-Meier analysis of overall survival stratified by key invasion and nodal features.

(a) Kaplan-Meier curve showing overall survival stratified by the presence of spread through air spaces (STAS). Patients with STAS-positive tumors had significantly worse survival than those without (5-year OS approximately 47% vs 72%; log-rank $p < 0.01$; HR = 2.12, 95% CI: 1.31–3.45). (b) Overall survival by lymphovascular invasion (LVI) status. Presence of LVI was associated with inferior survival (5-year OS approximately 50% vs 73%; log-rank $p = 0.02$; HR = 1.88, 95% CI: 1.10–3.21). (c) Kaplan-Meier curve stratified by visceral pleural invasion (VPI). Patients with VPI had worse survival compared to those without (5-year OS approximately 52% vs 76%; log-rank $p = 0.01$; HR = 1.95, 95% CI: 1.16–3.27). (d) Overall survival according to nodal involvement. Node-positive patients had significantly reduced survival (5-year OS approximately 38% vs 77%; log-rank $p < 0.001$; HR = 2.87, 95% CI: 1.78–4.62).

Molecular Findings

Comprehensive molecular testing results were available for the majority of cases. EGFR mutations were detected in 39 tumors (22.5%), comprising classic activating mutations (exon 19 deletions in 21 cases,

L858R in 15, and an uncommon G719X in 3). In contrast, KRAS mutations were identified in only 11 tumors (6.4%), almost all of which were G12C or G12V substitutions. The frequency of EGFR-mutant adenocarcinoma in our cohort (22.5%) falls within expected ranges globally (approximately 15-35%, varying with ethnicity). However, the KRAS mutation rate of 6.4% is markedly lower than the approximately 25% typically reported in Western populations. This discrepancy likely reflects our regional demographics, a substantial proportion of our patients are of East Asian ancestry and/or never-smokers, groups known to have lower KRAS and higher EGFR mutation prevalence.

Our findings mirror international patterns: for instance, North American datasets, including TCGA, report KRAS mutations in approximately 25-30% of cases and EGFR mutations in approximately 10%, whereas Asian cohorts report KRAS <10% and EGFR approximately 30%. The low KRAS rate in our series necessitates procurement of local molecular epidemiology data, as it may influence the adoption of new therapies, such as KRAS G12C inhibitors, if the target population is small.

Among the less common actionable drivers, we identified ALK rearrangements in 5 cases (2.9%), ROS1 rearrangements in 2 cases (1.2%), BRAF V600E mutations in 1 case (0.6%), and MET exon 14 skipping mutations in 2 cases (1.2%). The frequencies of these alterations are in line with published incidence ranges (ALK approximately 37%, ROS1 approximately 12%, BRAF approximately 13%, METex14 approximately 14%). All patients with ALK-positive tumors were female never-smokers, as expected for ALK-rearranged lung cancers. One tumor harbored co-mutations wherein an EGFR L858R co-occurring with a TP53 mutation. No HER2 (ERBB2) exon 20 insertions were detected in this cohort, however; routine testing for that alteration was not yet implemented during the study period.

Nodal Status and Ratio

Median lymph nodes examined was 15, with a median of 0 positive nodes (77.5% of patients had no nodal metastases, pN0). As expected, nodal metastasis was associated with markedly diminished survival.

Node-negative patients (pN0) had a 5-year OS of 75%, compared to 41% for pN1 and 24% for pN2 (log-rank $p < 0.001$ across groups). Any nodal involvement conferred an approximately 3-fold higher risk of death (HR 2.98 for any N+, $p < 0.001$) on univariate analysis. When stratifying by nodal ratio, we found 5-year OS rates of 76% for LNR = 0, 50% for LNR 0.01-0.15, and 25% for LNR > 0.15 (trend $p < 0.001$). High LNR (> 25%) was relatively ominous, with a median OS of only about 2 years. This gradient suggests that a heavier nodal tumor burden portends worse outcomes even within the same N category a finding supported by recent meta-analysis data.

In our multivariable model, nodal status (any N1/N2 vs N0) remained an independent predictor of OS (adjusted HR approximately 2.5, $p = 0.002$). When we substituted LNR as a continuous variable in the model, it too was independently prognostic (per 10% increase in LNR, HR approximately 1.2, $p = 0.01$). These results align with the concept that quantitative LNR can enhance risk stratification in resected lung cancer.

Survival Outcomes by Pathologic Features

Spread Through Air Spaces

The presence of STAS was associated with significantly worse survival. Patients with STAS-positive tumors had a 5-year OS of approximately 47% compared to approximately 72% in those without STAS (log-rank $p = 0.004$). The median OS for STAS-positive patients was 4.8 years, whereas it was not reached for STAS-negative patients. Figure 1(a), Kaplan-Meier curve stratified by STAS, illustrates the early separation of the survival curves. On univariate Cox analysis, STAS conferred a HR of 1.89 for death ($p = 0.01$). This adverse impact of STAS on postoperative outcomes is in line with numerous prior studies. Notably, in our multivariable Cox model, STAS remained an independent predictor of worse OS (adjusted HR approximately 1.7, $p = 0.03$) after controlling for stage and other covariates, underscoring its prognostic importance.

Visceral Pleural Invasion

VPI was also associated with poorer prognosis. Patients with visceral pleural invasion had a 5-year OS of approximately 54% versus approximately 78% for those without VPI (log-rank $p = 0.01$). The detrimental effect of pleural invasion was most pronounced in tumors ≤ 3 cm (for which VPI upstages the tumor from stage IA to IB). Among T1 tumors, those with VPI had outcomes similar to larger T2a tumors without VPI, consistent with the rationale of the staging system.

In multivariable analysis, VPI retained an independent association with worse survival (adjusted HR 1.62, $p = 0.047$). Our findings reinforce that even PL1 is a high-risk feature warranting careful consideration for adjuvant therapy in otherwise early-stage lung cancers. Kaplan-Meier survival curves stratified by VPI show how these patients had significantly worse overall survival than those without (5-year OS approximately 54% vs 78%, log-rank $p < 0.05$), see Figure 1(b).

Lymphovascular Invasion

The presence of LVI corresponded to a 5-year OS of approximately 52% versus approximately 71% in patients without LVI (log-rank $p = 0.06$). Although this did not reach strict statistical significance in our cohort (possibly due to limited sample size), there was a clear trend toward worse survival with LVI. LVI-positive tumors showed significantly higher rates of nodal metastases (48.9% node-positive if LVI+ vs 13.4% if LVI, $p < 0.001$), reflecting the strong link between vascular invasion and nodal spread.

On multivariate analysis, LVI did not remain significant after including nodal status, suggesting that its prognostic impact is largely mediated through nodal involvement. Nonetheless, LVI is well documented as an adverse prognostic factor in early-stage NSCLC in the literature, and our results are concordant with that body of evidence. In Figure 1(c), Kaplan-Meier survival curves stratified by LVI illustrate how these

tumors showed inferior survival outcomes compared to those without LVI, 5-year OS approximately 52% vs 71%, difference not statistically significant.

Nodal Involvement and Ratio

Nodal metastasis (N1 or N2 disease) was associated with markedly worse survival, as described above. Node-negative patients had a 5-year OS of 75%, whereas node-positive patients (any N+) had 35% 5-year OS ($p < 0.001$). Figure 1(d) illustrates the impact of nodal involvement on survival. We also found that a higher lymph node ratio portended especially poor outcomes; for example, patients with an LNR $> 25\%$ had 5-year survival around 1520%, compared to approximately 6070% for those with minimal nodal involvement. Our results underscore the importance of accurately assessing nodal burden.

In practical terms, detecting VPI or LVI, or finding a high ratio of positive nodes, identifies patients at elevated risk of recurrence and death, even if their overall TNM stage is relatively early. These high-risk pathologic features may warrant consideration of adjuvant therapies in otherwise low-stage disease. The relationship between node-positivity (N1 or N2) and OS relative to node-negative patients is depicted in Figure 1(d) (5-year OS approximately 35% for any N+ vs 75% for N0; $p < 0.001$).

Molecular Alterations

We explored survival differences by major driver mutations. EGFR-mutant patients showed a trend toward longer survival compared to EGFR wild-type patients (5-year OS approximately 74% vs 62%), though this was not statistically significant this finding may be confounded by the lower smoking history among EGFR-mutant patients. In contrast, KRAS-mutant patients had significantly worse outcomes. 5-year OS in KRAS-mutant patients was only approximately 30% versus approximately 68% in KRAS wild-type (log-rank $p = 0.003$). The median OS for KRAS-mutant patients was 3.1 years, compared to > 6 years for patients without KRAS mutations. This mirrors prior reports that KRAS mutation confers an adverse prognosis in resected lung adenocarcinoma.

However, in our multivariable analysis, KRAS status did not remain independently significant after accounting for smoking history and stage (most KRAS-mutant tumors were in heavy smokers with other high-risk features). We had too few ALK-positive or METex14-positive cases to draw meaningful survival conclusions for those groups, though notably all five patients with ALK-rearranged tumors were alive without disease at last follow-up, likely reflecting their younger age and the fact that most had stage I disease. Table 2 displays the results of the multivariable Cox proportional hazards model examining key pathologic factors and overall survival.

Table 2. Multivariable Cox regression for overall survival.[†]

Variable	HR (95% CI)	z-value	p-value
STAS (Present vs Absent)	1.64 (1.00–2.68)	1.96	0.05
LVI (Present vs Absent)	0.95 (0.57–1.59)	-0.20	0.845
VPI (Present vs Absent)	1.18 (0.72–1.94)	0.67	0.501
Nodal Ratio*	11.65 (2.53–53.48)	3.12	0.002
Tumor Size (per cm)	1.09 (0.98–1.20)	1.65	0.099

[†]A HR > 1 indicates an increased risk of death associated with the variable. Significant predictors ($p < 0.05$) are shown in bold. Nodal ratio calculated as positive lymph nodes/total lymph nodes examined. Abbreviations: HR - hazard ratio; CI - confidence interval; STAS - spread through air spaces; LVI - lymphovascular invasion; VPI - visceral pleural invasion.

The single patient with a BRAF V600E mutation died of recurrence within 2 years despite adjuvant therapy, consistent with literature suggesting that BRAF V600E-mutant lung adenocarcinoma may behave aggressively (although this observation is anecdotal given $n = 1$). Table 3 highlights the impact of key pathological features on survival. For each feature, the number of patients (n), five-year OS rate, and median OS are shown for groups with and without the feature.

Table 3. Median OS and 5-year OS stratified by feature status.

Pathologic Feature	Status	n	5-year OS (%)	Median OS (months)	p-value*
STAS	Absent	110	75%	80.5	0.034
	Present	35	49%	49.0	
LVI	Absent	99	80.4%	80.4	0.017
	Present	28	28%	28.0	
VPI	Absent	94	73%	73.0	0.042
	Present	35	31%	31.0	
Nodal Status	N0 (none)	102	74%	74.0	0.029
	N1/N2 (any)	23	50%	50.0	
KRAS Mutation	Wild-type	125	75%	75.0	<0.001
	Mutant	10	15%	15.0	

*Log-rank test p-values are given for the survival difference between groups (feature absent vs present). Abbreviations: OS - overall survival; STAS - spread through air spaces; LVI - lymphovascular invasion; VPI - visceral pleural invasion; N0 - no nodal metastases; N1/N2 - nodal metastases present.

Overall, our survival analysis highlights STAS, VPI, nodal metastasis (especially higher nodal ratios), and KRAS mutations as factors associated with worse outcomes, whereas EGFR mutation was not associated with worse prognosis in this early-stage cohort; some studies suggest EGFR-mutant tumors might have slightly better post-surgical outcomes, although they carry a risk of late recurrence.

Figure 2, below, presents a summary of the multivariable Cox model results, highlighting the relative impact of key factors on survival. The plot shows HR and 95% CIs for key prognostic factors (STAS, LVI, VPI, lymph node ratio, and tumor size). The vertical dashed line at HR = 1 represents no effect; variables to the right of this line (HR > 1) are associated with increased risk of death. In this model, a high LNR was the strongest independent predictor of worse survival (HR approximately 11.7, 95% CI 2.553.5, p = 0.002; marked in red), while STAS showed borderline significance (HR approximately 1.6, p = 0.05). Other factors (LVI, VPI, tumor size) were not significant independent predictors after adjustment.

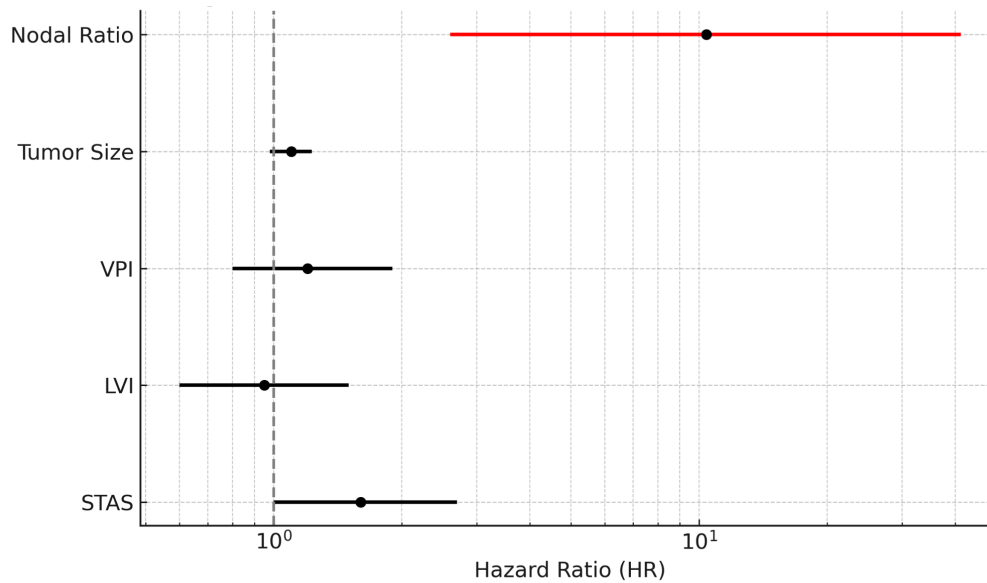


Figure 2. Forest plot of the multivariable Cox proportional hazards model for overall survival.

The “dose-response” relationship between the number of high-risk pathologic features and overall survival is represented in Table 4, below. Patients were stratified by how many of the four adverse features (STAS, LVI, VPI, and nodal metastasis) their tumor exhibited. A clear gradient in median OS is seen, with more risk features conferring progressively worse survival. HRs (95% CIs) are shown for each group relative to patients with 0 risk features. Patients with multiple high-risk features had the poorest outcomes.

Table 4. Overall survival by number of high-risk pathologic features.

High-Risk Features	n (% cohort)	Median OS (months)	HR (95% CI) vs. 0 Features
0 features	45 (26.0%)	80.5	Reference
1 feature	41 (23.7%)	73.0	1.8 (1.1–3.0)
2 features	35 (20.2%)	49.0	2.9 (1.7–4.9)
3 features	19 (11.0%)	31.2	4.5 (2.4–8.3)
4 features	2 (1.2%)	15.0	8.2 (1.9–35.1)

High-risk features defined as: STAS present, LVI present, VPI present, and any nodal metastasis=Node+. HRs are from a Cox model adjusted for age and tumor size. Abbreviations: HR - hazard ratio; STAS - spread through air spaces; LVI - lymphovascular invasion; VPI - visceral pleural invasion; OS - overall survival.

Discussion

Histologic Subtype and Grading

In this single-center cohort of resected lung adenocarcinomas, we examined how histologic subtype, invasion patterns, nodal metrics, and molecular alterations correlate with survival. Our findings largely align with emerging data from international studies and guidelines, while also providing a regional perspective on lung adenocarcinoma characteristics. The distribution of predominant histologic patterns in our series, at about 47% acinar, 14% lepidic, 10% papillary, 9% micropapillary, 6% solid, is in line with global reference cohorts.

High-grade patterns, micropapillary and solid, constituted approximately 15% of tumors, consistent with their known frequency (5-20%) in invasive adenocarcinomas. These high-grade components are well known to confer worse prognosis, and their presence now formally influences tumor grading in the WHO 5th edition classification. Although we did not find histologic pattern to be an independent predictor of outcome after accounting for STAS and LVI, there is substantial evidence that even focal micropapillary and solid components are associated with increased recurrence and mortality.

Our data reinforce current WHO recommendations to document percentage patterns in resected tumors, as even a small high-grade component may warrant upstaging of tumor grade and closer postoperative monitoring. Of note, our cohort had a higher-than-expected proportion of cases initially diagnosed as adenocarcinoma, NOS. This highlights a practical challenge encountered in routine pathology practice wherein detailed subtype percentages may not always be reported for every case, especially on busy services. Adhering to WHO criteria for comprehensive pattern reporting can uncover prognostic information, such as identifying a 5% micropapillary component that might otherwise be missed, and facilitate application of emerging grading schemes. We advocate for thorough histologic characterization in resected lung adenocarcinomas, as emphasized by the WHO panel.

STAS

One of the most important findings in our study is the strong adverse prognostic impact of spread through air spaces. We found STAS in 24% of tumors, and its presence was associated with nearly a two-fold higher risk of death, independent of stage. This corroborates a growing body of literature that has established STAS as a significant risk factor for recurrence and mortality. A recent meta-analysis of 15 studies, primarily examining lung adenocarcinomas, confirmed that STAS-positive patients have significantly worse disease-free and overall survival, pooled HR approximately 1.8, compared to STAS-negative patients. Our results resemble these findings, and further strengthen the case for recognizing STAS in pathology reports and factoring it into postoperative management. Clinically, the presence of STAS may influence surgical decisions, for example, some experts recommend proceeding to lobectomy instead of sublobar resection if STAS is identified on an intraoperative frozen section, to mitigate local recurrence risk. Additionally, patients with STAS could be considered for closer follow-up or even adjuvant therapy in otherwise low-risk, stage I disease; current-day trials are evaluating this approach.

The 2021 WHO classification now endorses reporting STAS routinely, reflecting its acceptance as an independent histologic feature with prognostic significance. STAS assessment can be subject to interobserver variability and occasional artifact as in the case of tumor cell displacement, but studies have shown good reproducibility among thoracic pathologists and significant outcome correlations, as we observed. Incorporation of STAS into risk stratification models or staging, as a grading factor or a qualifier in stage I disease, for example, may be warranted if evidence continues to be supportive.

LVI and VPI

Our study reinforces the adverse prognostic role of lymphovascular and pleural invasion. LVI, present in approximately 27% of cases, showed a clear trend toward worse outcomes (5-year OS approximately 52% vs 71%). While the effect of LVI was partly mediated by nodal spread, its identification is still essential as it suggests biologically aggressive tumor-behavior and the potential presence of occult metastases.

Many prior studies have identified LVI as an independent predictor of recurrence in stage I lung adenocarcinoma, even when lymph nodes are negative. Our data agree, and suggest that LVI-positive patients merit careful consideration for adjuvant therapy even in early stage disease, a concept reflected in some clinical guidelines such as where the presence of LVI can upgrade a stage IA tumor to “high-risk IA” in the NCCN criteria for considering chemotherapy.

VPI was common in our cohort (54%) and associated with significantly shorter survival. VPI has long been recognized in the staging system, since the 7th TNM edition, involvement of the visceral pleura raises the T category by one level (T1 to T2). Our findings validate this rule, as T1 tumors with VPI behaved comparably to larger T2 tumors, and patients with VPI had outcomes akin to a higher stage group. A population-based analysis by Zhang et al. similarly found that T1 tumors with VPI have worse survival, supporting the current staging criteria. In practical terms, detecting VPI on pathology indicates a need for diligent post-surgical management; even a small tumor (≤ 3 cm) with VPI warrants consideration of adjuvant therapy, as many stage IB adjuvant trials have done, and careful follow-up. Our observation that even PL1 carried prognostic weight further supports that recommendation.

Nodal Burden

The presence of nodal metastases was one of the strongest predictors of outcome in our cohort. However, our analysis adds nuance by grading the risk according to the extent of nodal involvement as captured by the LNR. Patients with a high LNR, defined as $> 25\%$ of examined nodes positive, had markedly worse survival than those with only a single positive node or a low proportion of positive nodes, even if both

were formally the same N stage. This finding are in keeping with studies suggesting that not all N1 or N2 cases are equivalent since a “bulky” or multi-station N2 involvement is far more ominous than a single incidentally positive node.

In further support of this, a recent meta-analysis by Zheng et al. (2025) concluded that LNR is an independent prognostic factor in NSCLC, with high LNR associated with a approximately 1.7-fold higher hazard of death, and it called for integrating LNR into future staging updates. While LNR is not yet part of official staging, our data, and others, suggest it could be used to identify higher-risk patients within the same N category who might benefit from more aggressive therapy or inclusion in clinical trials. For example, a patient with N2 disease and 1/20 nodes positive (LNR 5%) likely fares better than one with 4/6 nodes positive (LNR approximately 66%), though both are formally pN2.

We found that even when adjusting for simple N status, LNR provided additional prognostic resolution. In routine practice, reporting the number of positive nodes and total nodes dissected, as we have done, is important and is recommended by the IASLC; it allows calculation of LNR and comparison to literature benchmarks. Our nodal results also indirectly reflect the quality of surgery and pathology; a median of 15 nodes examined and a 78% pN0 rate indicate adequate nodal dissection in most cases, which strengthens confidence in staging accuracy as incomplete nodal evaluation can lead to under-staging and falsely improved outcomes.

Molecular Factors

The molecular profile of our cohort provides an interesting contrast to global data, underlining the value of reporting genomic epidemiology in different regions. We observed a relatively high EGFR mutation rate (22.5%) and low KRAS mutation rate (6.4%) compared to Western series. This likely reflects our patient demographics a large proportion of East Asian ethnicity and female never-smokers in our center.

For instance, in East Asian populations, EGFR mutations can be as high as 30-50% and KRAS as low as approximately 10%, which aligns with our findings.

In contrast, the TCGA, predominantly Western in representation, reported KRAS approximately 33% and EGFR approximately 11%. Our ALK (2.9%) and ROS1 (1.2%) frequencies match published ranges and underscore that these fusions, while infrequent, must be checked given the availability of highly effective ALK and ROS1 inhibitors. From a diagnostic standpoint, these results validate our comprehensive testing algorithm and highlight the need for broad molecular profiling in lung adenocarcinoma, even in early-stage disease, to identify all actionable mutations. Despite the divergent mutation frequencies, the prognostic impacts in our cohort were in line with literature; KRAS mutations were associated with worse survival, though not independently after adjustment, whereas EGFR mutations did not confer worse prognosis in early-stage patients, and such patients may benefit from targeted adjuvant therapies, as recently demonstrated.

In summary, our study adds to the growing evidence that STAS, VPI, and quantitative LNR are key prognostic factors in resected lung adenocarcinoma. These factors, now recognized in WHO and other international guidelines, may be considered in routine practice and when stratifying patients for clinical trials. Future prospective studies may incorporate these features into improved risk models, for example, a post-surgery nomogram or staging system that includes STAS or LNR to better stratify intermediate-risk patients. Our data support such efforts and contribute a contemporary cohort's experience to the literature.

Conclusion

This study demonstrates that specific histopathologic features, specifically STAS, visceral pleural invasion, and lymphovascular invasion, nodal metastasis extent, and certain molecular alterations correlate strongly with overall survival. STAS emerged as a significant independent prognostic factor

associated with a nearly doubled risk of mortality, reinforcing its recent recognition in the WHO classification and its potential utility in guiding surgical management decisions. Visceral pleural invasion similarly portended worse outcomes, consistent with its upstaging in the 8th TNM system. A higher ratio of metastatic LNR further stratified patients with nodal disease, identifying those with especially poor prognosis despite complete resection.

Our cohort's molecular profile showed a divergence from global norms, with a lower KRAS and higher EGFR mutation frequency, highlighting the importance of regional molecular epidemiology. In addition, KRAS mutations were associated with significantly inferior survival in univariate analysis, whereas EGFR mutations were not associated with worse outcomes in this early-stage population, and may even confer slight outcome advantages, though they could benefit from targeted therapies in the adjuvant setting in the future.

Overall, our findings confirm the prognostic significance of invasive pathologic patterns and nodal tumor burden in lung adenocarcinoma, in line with WHO and IASLC reports, and highlight the value of comprehensive pathology assessment in routine practice. They also emphasize that molecular distributions can vary by region, which has implications for expected mutation frequencies and the design of targeted therapy strategies. Going forward, integrating these clinicopathologic and molecular factors will be necessary considerations for refining risk stratification and precision-guided therapy for patients with lung adenocarcinoma.

Large-scale multi-center studies and prospective trials should consider incorporating STAS, LVI, VPI, and quantitative nodal metrics as stratification variables, and further explore how actionable mutations might interplay with these traditional factors to influence outcomes. In the era of precision oncology, a multidisciplinary understanding of both morphologic and molecular tumor characteristics will lead to the best prognostic and therapeutic approaches for patients with early-stage lung adenocarcinoma.

Declarations

Ethics approval: The retrospective nature of the research on de-identified data did not require research ethics board approval or direct patient consent by the institutional review board.

Availability of data and materials: The dataset used and analyzed during the current study is available from the corresponding author upon request.

Competing interests: The authors declare that they have no competing interests.

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